

THE LIMINAL INITIATIVE · LIMINAL MATH

The Liminal Career Atlas

Where mathematics takes you: careers for
mathematics graduates and olympiad medallists



- 16 careers explained
- What an olympiad medal buys
- Pay data for India and abroad
- A plan by stage

INSIDE THIS ATLAS

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INTRODUCTION

Before you choose

A mathematics degree is a licence, not a job title

During the 2024–25 placement season, a computer science student at IIT Madras accepted a pre-placement offer from the trading firm Jane Street reported at ₹4.3 crore a year — salary, bonus and relocation included — for a role in Hong Kong. The figure made headlines for a week. What the headlines missed was the more useful fact underneath it: the firm making that offer does not hire for a branch, a degree title or a college name. It hires for one thing, and it tests for that thing with puzzles a strong olympiad student would recognise on sight.

That quiet shift is what this atlas is about. For most of the last century, the honest answer to “*what does a mathematics student become?*” was short — a teacher, a professor, perhaps a statistician in a government office. In 2026 the answer is long, and parts of it are startling. Mathematics now sits underneath trading desks, AI laboratories, cryptographic standards, insurance pricing, clinical trials, logistics networks and the software running on every phone in the country.

So the first idea to unlearn is that a mathematics degree is a job title. It is not. It is a licence — proof that you can enter an unfamiliar formal system, learn its rules faster than most people, and reason inside it without a guide. Employers are not buying the theorems you memorised. They are buying the speed at which you will learn *their* problem.

The conventional wisdom in many Indian households runs the other way: pure mathematics is the narrow, risky choice; engineering is the broad, safe one. An honest answer has to face the strongest version of that worry, and it is a real one. A BSc in mathematics, on its own, attracts fewer campus recruiters than a B.Tech from a comparable institution, and a student who stops at “*I am good at maths*” will find the market politely uninterested. But the conclusion people draw from that — avoid mathematics — is close to the opposite of the right one. Skills tied to today’s tools depreciate when the tools change, and this decade has shown how fast that can happen. Mathematics is the one asset in the portfolio that does not.

What the market actually rewards is **mathematics plus one**: mathematics plus programming, plus statistics, plus finance, plus the ability to explain. Almost every page that follows lands on that formula. The careers differ only in which “one” they demand.

How this atlas is built. Part I looks at what an olympiad medal actually buys — in admissions, in hiring and over a lifetime — and what it does not. Part II gives each of sixteen careers its own page-set, built the same way so they can be lifted straight onto a website: an at-a-glance panel, what the job really is, the mathematics it uses, how to get in from India, where an olympiad background helps, the honest catch, and what a student can start this year. Part III turns the atlas into a plan.

A note on the numbers. US figures are median annual wages from the US Bureau of Labor Statistics (Occupational Employment and Wage Statistics, May 2025). They are base pay and exclude bonus and equity, which matter enormously in finance and technology. Indian figures come from official notifications, campus-placement reporting and salary aggregators, and should be read as indicative ranges, not promises. The eye-catching quant numbers are concentrated in a handful of firms and campuses. Everything here was checked in October 2026; admission schemes and pay change every year, so verify against the official source before you decide.

PART I

What a medal buys — and what it doesn't

Admissions, employers and a lifetime of evidence — and an honest account of what a medal cannot do for you.



PART I · THE OLYMPIAD ADVANTAGE

What a medal buys — and what it doesn't

The ladder

In India, olympiad mathematics is a five-rung ladder run by the Homi Bhabha Centre for Science Education (HBCSE), a national centre of the Tata Institute of Fundamental Research: the IOQM (Indian Olympiad Qualifier in Mathematics), then the RMO (Regional Mathematical Olympiad), the INMO (Indian National Mathematical Olympiad), the IMO Training Camp, and finally a six-member team at the International Mathematical Olympiad.

The top of that ladder has rarely been stronger. India finished 4th in the world in 2024 — its best ever — with four gold medals. In 2025 it came 7th with three golds and a record team score of 193 out of 252. At Shanghai in 2026, the largest IMO ever held with around 666 contestants, all six Indian students medalled (two gold, four silver) and the team again finished 7th. Of India's 25 IMO golds since its 1989 debut, 14 have come in the last seven editions.

That pipeline is now one of the most reliable talent signals in the country. Institutions and employers have noticed.

Door one: admissions

The first thing a strong olympiad record buys is a way around India's entrance-exam bottleneck. None of these routes is automatic, and most still involve a screening or an interview — but each replaces a lottery of marks with a judgement of ability.

Institution	Route	What it requires (check each year)
CMI, Chennai	Direct admission to BS (Hons)	Qualifying for the IMO or EGMO Training Camp in Class 11 or 12 (Physics OCSC also qualifies)
ISI	Written test waived for B.Math / B.Stat	INMO awardee status; the interview still applies
IIT Bombay	BS Mathematics outside JEE	INMO / IMO-route qualifiers; departmental screening
IIT Kanpur	Olympiad admissions (since 2025–26)	International-olympiad-level record; five departments incl. Mathematics & Statistics and CSE; 14 students admitted for 2026–27
IIT Madras	ScOpE (since 2025–26)	Two supernumerary seats per UG programme, one reserved for female candidates; olympiad-camp record
IIT Kharagpur	SCOPE (from 2026–27)	Olympiad achievers; supernumerary seats that do not reduce JEE intake

Schemes change every year; supernumerary seats are added on top of regular intake. Always confirm on the institute's official admissions page.

Door two: employers

The interview at a top trading firm is not a finance exam. It is an olympiad round with money in the examples. That is not a figure of speech: Jane Street, Citadel and OpenAI are the IMO's main sponsors, Citadel takes on dozens of olympiad competitors as interns every year, and many of Hudson River Trading's earliest interns already knew each other as teenagers from the olympiad circuit. Quant firms now send people to high-school mathematics tournaments the way football clubs send scouts to academy matches.

Count where medallists end up and the picture sharpens. One tally of firms employing international olympiad medallists placed Jane Street fourth — behind only Google, Meta and Google DeepMind. Read that list again: two AI labs, a technology giant and a trading firm. The modern market for mathematical talent is, in large part, a contest between finance and artificial intelligence.

Door three: a lifetime

The best long-run evidence comes from economists Ruchir Agarwal and Patrick Gaule, who traced 4,710 IMO participants from 1981 to 2000. Around 22% went on to a mathematics PhD. Each additional point scored as a teenager (out of 42) was associated with 2.6% more mathematics publications and 4.5% more citations over a lifetime. And an IMO gold medallist was roughly fifty times more likely to win a Fields Medal than a PhD graduate of a top-ten mathematics department.

The finding that matters most for India sits in the second half of their paper. Participants from low-income countries produced about 35% fewer mathematics publications and over 50% fewer citations than equally talented participants from rich countries — and the gap came largely from not going on to a PhD at all. The penalty was not talent. It was access. Strip away the medals and this is a market-design problem: ability is spread fairly evenly across the world; the routes out of it are not.

What a medal does not buy

Here the atlas has to be honest, because parents rarely are. Roughly four in five IMO participants never earn a mathematics PhD. Contest mathematics and research mathematics are different sports: an olympiad problem is designed to have a clean solution reachable in ninety minutes; a research problem may have no solution, and nobody knows in advance. A medal is a signal with an expiry date — it opens the first interview and the first admission, and by your second job nobody asks.

And most students who climb the ladder never reach the camp at the top. That is fine. The habits the climb builds — patience with a hard problem, proof as a standard of honesty, the instinct to test a claim before trusting it — transfer to every career in this atlas. The medal is the receipt. The training is the purchase.

A medal opens the first door. It has never once walked anyone through the second.

OVERVIEW

The landscape at a glance

Sixteen careers, one table. The US column is a benchmark, not a forecast for India; the Indian column uses verified public figures where they exist and is left descriptive where they do not. “Olympiad edge” is our judgement of how much contest training helps you get in.

#	Career	Typical entry	India: indicative pay	US median (BLS, May 2025)	Olympiad edge
01	Quantitative Researcher	BS / MS / PhD	Top campus offers ₹90 L+	Not tracked separately	Very high
02	Quantitative Trader	BS, often straight from college	Same tier as researchers	Not tracked separately	Very high
03	Quant Developer	BS in CS / maths	Top HFT offers ~₹80–90 L	\$135,980 (software developers)	High
04	Research Mathematician	PhD + postdocs	PhD stipend ₹37–42 k / month	\$126,710 (mathematicians)	High
05	AI / ML Research Scientist	MS / PhD common	Wide; global labs in Bengaluru	\$140,300 (research scientists)	High
06	Formal Mathematics (Lean)	Degree + public proofs	Emerging; mostly global roles	Not tracked	High
07	Data Scientist	Any quantitative degree	Freshers ~₹6–9 LPA	\$120,230	Medium
08	Actuary	ACET + exams	₹4–8 LPA early; Fellows far higher	\$130,000	Medium
09	Cryptographer	MS / PhD for research	Wide	\$129,180 (info-security)	High
10	Software Engineer	BS	Wide; product firms highest	\$135,980	Medium–high
11	Statistician / Biostatistician	MSc / M.Stat; UPSC ISS	ISS: Level 10, ₹56,100 basic	\$105,650 (statisticians)	Medium
12	Operations Research Analyst	BS / MS	Analytics & logistics roles	\$88,940	Medium
13	Economist	MA; PhD for research	IES: Level 10 start	\$124,720	Medium
14	Risk & Financial Engineering	MSc / MFE / FRM	Bank analytics centres	Not tracked separately	Medium
15	Educator & Olympiad Coach	BSc / MSc / B.Ed	Very wide	—	Very high
16	Founder	A problem and a team	Equity, usually nothing at first	—	Medium

PART II

Sixteen careers

One page-set per career, each built the same way: what the job really is, the mathematics it uses, how to get in from India, the olympiad edge, the catch, and what to start this year.



CAREER 01 / 16

Quantitative Researcher

Finding signal in a market that is almost all noise



AT A GLANCE

TYPICAL ENTRY	BS, MS or PhD in mathematics, statistics, CS or physics
CORE MATHEMATICS	Probability, statistics, linear algebra, optimisation, time series
PAY — INDIA	Top campus offers reported around ₹90 lakh+ including bonus; Quadeye lists a minimum ₹93 LPA on internship conversion
PAY — ABROAD	Among the highest new-graduate pay in any field; not tracked separately by BLS
OLYMPIAD EDGE	Very high at the interview; fades quickly after it
WHERE	Hedge funds, proprietary trading firms, market makers — Mumbai, Gurugram, Bengaluru, Singapore, Hong Kong, London, New York

What the job really is

A quantitative researcher spends most of the year being wrong. Hundreds of ideas — a pattern in order-book data, a lag between a news headline and a price, a relationship between two currencies — die in testing. The few that survive become strategies that trade thousands of times a day. The job is not predicting the market. It is building statistical evidence strong enough to bet money on, and staying humble enough to abandon it when it decays.

That makes the work closer to experimental science than to finance as most people picture it: a hypothesis, a dataset, a test, and — usually — a rejection. Researchers hand the survivors to traders and developers who run them live.

The mathematics you will actually use

Probability and statistics above everything: regression, time-series models, hypothesis testing, and the subtle trap of *multiple testing* — if you test a thousand ideas, some will look brilliant by pure chance. Linear algebra for covariance matrices and factor models, optimisation for building portfolios, stochastic processes on derivatives desks, and increasingly machine learning. Daily tools are Python and often C++.

How to get in from India

The funnel runs mainly through campus placements at the IITs, BITS, ISI, CMI and IISc. The field has become intensely competitive: global firms such as Jane Street, Hudson River Trading, Optiver and IMC now compete in India against home-grown firms like Quadeye and Graviton, and several have opened Indian offices in Mumbai. At IIT Kharagpur alone, students reported 38 quant offers in one recent internship season, up from about 20 the year before.

Interviews are puzzle-heavy: probability brainteasers, expected-value questions, mental arithmetic, and a coding round. Abroad, a master's in financial engineering or a PhD in maths, physics or statistics is a common route into research roles.

Where the olympiad helps — and where it stops

The interview is, in effect, a combinatorics and probability paper under time pressure, so contest training is a genuine advantage. After the interview, the advantage fades. What keeps a researcher employed is statistical discipline — distrusting a beautiful backtest — not cleverness.

The catch

Your output is measured in P&L (profit and loss), and strategies decay as competitors discover them. Secrecy means you cannot publish or talk about your best work. Hours can be long, and the headline pay is bonus-heavy and concentrated at a few firms.

START THIS YEAR

- 1 Learn probability properly, not just formulas — work through a rigorous undergraduate text end to end.
- 2 Get fluent in Python and pandas; download any public market dataset and try to break your own ideas.
- 3 Practise expected-value and conditional-probability puzzles weekly; aim for a quant internship in your third year.

In research, being clever gets you the interview. Being suspicious of your own results keeps you the job.

CAREER 02 / 16

Quantitative Trader

Decisions under uncertainty, at speed, with real money



AT A GLANCE

TYPICAL ENTRY	BS in mathematics, CS or physics; many firms hire straight from undergraduate
CORE MATHEMATICS	Expected value, conditional probability, Bayesian updating, game theory, options mathematics, mental arithmetic
PAY — INDIA	Same tier as quant research; heavily bonus-driven
PAY — ABROAD	Not tracked separately by BLS; top-decile graduate pay
OLYMPIAD EDGE	Very high
WHERE	Market makers and proprietary trading firms — Optiver, IMC, SIG, Jane Street, HRT; Indian firms such as Graviton and Quadeye

What the job really is

If the researcher is the scientist, the trader is the pilot. Market makers quote prices to buy and sell thousands of instruments continuously, earning a small edge on each trade. The trader runs that machine: setting risk limits, adjusting prices when the world changes, and — the hardest part — deciding when the model is wrong and a human should step in.

Trading at these firms is often imagined as gambling in a nicer shirt. It is closer to the opposite. It is the business of refusing bets with negative expected value, thousands of times a day, and taking positive ones in the right size.

The mathematics you will actually use

Expected value and conditional probability, all day. *Bayesian updating* — revising a belief in proportion to new evidence. Game theory, because the other side of every trade is a thinking opponent. Options mathematics, including the “Greeks” (the sensitivities of an option’s price to the underlying price, time and volatility). And fast, accurate mental arithmetic. It is less research-heavy than quant research and far more about real-time judgement.

How to get in from India

This is one of the few elite finance careers open directly to strong undergraduates. Firms such as Jane Street, Optiver and HRT hire many people straight from their bachelor’s degree, and interviews lean on probability, mental maths and market-making games rather than finance theory. Selectivity is extreme — published estimates put acceptance at top firms below 2% of applicants.

Where the olympiad helps

Almost everywhere in the hiring process. These are the firms that sponsor the IMO and recruit at olympiad events. A medal tells them you can think clearly under a clock — exactly what the trading floor demands.

The catch

Your results are visible daily, sometimes hourly, and stress is part of the job description. The skill set is specialised: excellent trading judgement does not always transfer to other careers, and screen time is relentless.

START THIS YEAR

- 1 Train mental arithmetic with timed drills; speed and accuracy are both tested.
- 2 Work through a standard quant-interview problem book, writing full solutions, not just answers.
- 3 Learn the basics of options pricing and why market makers hedge.

The market does not pay you for being right. It pays you for knowing exactly how right you are.

CAREER 03 / 16

Quant Developer & HFT Engineer

Where research becomes orders, measured in microseconds



AT A GLANCE

TYPICAL ENTRY	BS in CS or mathematics with strong programming
CORE MATHEMATICS	Algorithms, data structures, discrete mathematics, probability, numerical methods
PAY — INDIA	Top high-frequency trading campus offers reported around ₹80–90 lakh
PAY — ABROAD	US software developers: median \$135,980 (BLS, May 2025); trading firms pay well above
OLYMPIAD EDGE	High — especially the informatics olympiad and competitive programming
WHERE	HFT firms, market makers, hedge-fund technology teams

What the job really is

In high-frequency trading, a microsecond — one millionth of a second — can be the whole competitive advantage. Quant developers build the systems where research turns into orders: market-data handlers, execution engines, risk checks and the simulators researchers use to test ideas.

This is often described as “just engineering at a finance firm.” It isn’t. The developer’s code *is* the strategy in production, and an idea that is right but slow is, in this business, simply wrong.

The mathematics you will actually use

Algorithmic complexity and data structures, discrete mathematics, probability for simulation, and numerical methods. Languages lean low-level — modern C++, Linux internals, networking — with Python for tooling, and occasionally functional languages (Jane Street famously runs on OCaml).

How to get in from India

Campus placements at top institutes, plus a visible record in competitive programming — Codeforces ratings, ICPC results, the Indian National Olympiad in Informatics. A mathematics degree with a computer science minor (CMI’s Mathematics and Computer Science track is a good example) is a strong base if you can prove you build, not just solve.

Where the olympiad helps

The informatics olympiad is the closest analogue to this job’s interviews. Mathematical olympiad training helps with the algorithm rounds and with reasoning about edge cases — the bugs that cost real money.

The catch

Production pressure and on-call duty, a narrow and specialised technology stack, and the knowledge that a single defect can be very expensive in a very short time.

START THIS YEAR

- 1 Pick C++ or Rust and learn how memory and caches actually work, not just syntax.
- 2 Compete regularly on a competitive-programming platform; consistency beats bursts.
- 3 Build a small order-book simulator — it teaches more than any course.

Speed is not a feature of the strategy. In this corner of finance, speed is the strategy.

CAREER 04 / 16

Research Mathematician

The only job on this list where the question is yours



AT A GLANCE

TYPICAL ENTRY	PhD, then one to three postdoctoral positions
CORE MATHEMATICS	Your field — algebraic geometry, number theory, analysis, combinatorics, probability, topology
PAY — INDIA	PhD stipend ₹37,000 a month for two years, then ₹42,000 (NBHM or CSIR-JRF), plus contingency grant; faculty on 7th CPC academic pay levels
PAY — ABROAD	US mathematicians: median \$126,710 (BLS, May 2025) — mostly government and industry roles
OLYMPIAD EDGE	High but partial
TIMELINE	Typically 8–12 years from Class 12 to a permanent post

What the job really is

Research mathematicians create new mathematics: they prove theorems nobody has proved, build theories that connect fields, and train the next generation. The day is a mix of thinking, reading, writing papers, refereeing other people's, teaching and supervising.

The key skill is not solving. It is asking — finding a question hard enough to matter and soft enough to crack. Many brilliant problem-solvers discover this in their first year of a PhD, and it can be a shock.

How to get in from India

A strong undergraduate base (CMI, ISI, the IISERs, IISc and others), then an MSc or integrated programme, then a PhD. The National Board for Higher Mathematics runs a written test each January; its doctoral scholarship pays ₹37,000 a month for the first two years and ₹42,000 thereafter, with a ₹40,000 annual contingency grant. CSIR-NET's Junior Research Fellowship pays at the same rates. TIFR, IMSc, ISI, CMI and HRI run their own entrance processes, and many students go abroad, where good PhD programmes are fully funded.

After the PhD comes the postdoc stage — usually two to six years of temporary positions, often in different countries — before a permanent faculty post at a research institute, IIT, IISER or university.

Where the olympiad helps — and where it doesn't

The Agarwal–Gaule data in Part I is clear that teenage olympiad performance predicts research output. But contest and research mathematics reward different temperaments. An olympiad problem has a known, elegant solution that fits in ninety minutes; a research problem may take months, may be false, and comes with no answer key. The transition from sprinting to marathon running is the single biggest hurdle for strong olympiad students who choose research.

The catch

Pay is the lowest on this list relative to ability. The postdoc years are insecure and geographically unsettled, permanent positions are scarce, and success depends on years of work with uncertain payoff.

START THIS YEAR

- 1 Read beyond the syllabus — one serious book outside your coursework each semester.
- 2 Apply to summer programmes such as NBHM's Mathematics Training and Talent Search (MTTS) and the Indian Academies' Summer Research Fellowships.
- 3 Write up proofs carefully, even for problems you have already solved; exposition is half of research.

Academia pays least in money and most in one currency nobody else offers: the right to choose your own question.

CAREER 05 / 16

AI & Machine Learning Research Scientist

Building the systems that are now learning mathematics themselves



AT A GLANCE

TYPICAL ENTRY	MS or PhD common for scientist roles; strong BS graduates enter as research engineers
CORE MATHEMATICS	Linear algebra, probability, optimisation, information theory, some analysis
PAY — INDIA	Wide range; global labs and AI startups in Bengaluru pay at the top of the tech market
PAY — ABROAD	US computer and information research scientists: median \$140,300, projected 22% growth 2025–35 (BLS)
OLYMPIAD EDGE	High
WHERE	AI labs, big-tech research groups, AI startups, quant firms' ML teams

What the job really is

In July 2025, AI systems from Google DeepMind and OpenAI reached gold-medal level on IMO problems, and Harmonic's Aristotle and ByteDance's Seed-Prover matched that with formally verified proofs. The interesting fact for a student is not that machines got good at olympiads. It is who built them: teams full of former olympiad contestants and mathematicians.

Research scientists design and analyse learning algorithms, run large experiments and publish; research engineers make those ideas work reliably at enormous scale. The line between the two is increasingly blurred.

The mathematics you will actually use

Linear algebra — almost everything in a neural network is a matrix operation. Probability and statistics, optimisation (gradient descent and its many descendants), information theory, and enough analysis to read theory papers. Alongside it: heavy programming in Python and deep-learning frameworks, and an understanding of distributed computing.

How to get in from India

Research publications at venues like NeurIPS, ICML and ICLR are the currency for scientist roles, which usually means a master's or PhD. Strong engineers can enter as research engineers with a bachelor's degree and a portfolio of serious projects. Bengaluru hosts research offices of several global technology companies, and the number of Indian AI startups is growing quickly.

Where the olympiad helps

Olympiad competitors are in demand at both AI labs and quant firms, and the two now compete for the same people. Contest training helps with the reasoning; the empirical, experiment-driven side of the work has to be learned separately.

The catch

The field moves weekly, so knowledge depreciates fast. Much of it is empirical: things work before anyone can prove why, which some mathematicians find uncomfortable. Hype is high, and frontier research is concentrated where the computing power is.

START THIS YEAR

- 1 Master linear algebra and probability at proof level, not just computation.
- 2 Reimplement a classic paper from scratch rather than calling a library.
- 3 Read one recent paper a week and try to reproduce one result per semester.

The machines have learned to solve olympiad problems. Someone still has to decide which problems are worth solving.

CAREER 06 / 16

Formal Mathematics & Proof Engineering

Lean, theorem provers, and the new economics of trust



AT A GLANCE	
TYPICAL ENTRY	Mathematics or CS degree plus a public portfolio of formal proofs
CORE MATHEMATICS	Logic, type theory, functional programming — plus the field you formalise
PAY — INDIA	Few dedicated roles yet; most opportunities are remote or abroad
PAY — ABROAD	An emerging category, not tracked by BLS
OLYMPIAD EDGE	High — rigour is the whole job
WHERE	AI-for-mathematics startups, the Lean community, verification teams at cloud and chip companies, academic formalisation projects

What the job really is

A proof assistant such as Lean is software that checks every logical step of a proof. Nothing is taken on trust; no “it is easy to see” survives. Every AI system that has reached medal level at the IMO with *formal* proofs has used Lean, and the tooling is maturing fast: the proof of Viazovska’s Fields-Medal-winning result on 24-dimensional sphere packing was recently autoformalised — over 200,000 lines of code — in about two weeks.

Formalisation is often dismissed as bookkeeping for proofs we already believe. It is closer to the opposite. It is what lets anyone — or anything — trust a proof they cannot personally check. As AI systems produce more mathematics than humans can read, that trust becomes the scarce good.

What people in this field do

Build and maintain shared libraries of formalised mathematics (Mathlib is the largest), design autoformalisation pipelines that translate ordinary mathematics into Lean, and verify software, hardware and cryptographic code where bugs are unacceptable. Startups such as Harmonic and Axiom work at the AI end — Axiom reported solving all twelve Putnam 2025 problems — while cloud providers run automated-reasoning groups that hire for exactly these skills.

How to get in from India

This is a field where your work is your CV. Contribute to Mathlib, join the Lean community’s online discussions, and formalise something real. A master’s or PhD in logic, programming languages or mathematics strengthens research roles. It is also why Lean-first mathematics education matters: students who learn rigour as something a machine checks build a habit that the next decade will reward.

The catch

The field is small and its job market is still taking shape. The tools can be painful, and formalising a single page of mathematics can take days. It is an early bet — with the risks that early bets carry.

START THIS YEAR

- 1 Play the Natural Number Game, then work through *Mathematics in Lean*.
- 2 Formalise one olympiad problem you have already solved on paper — and notice every step you skipped.
- 3 Make a first small contribution to Mathlib and get it reviewed.

When machines can produce proofs faster than people can read them, the rarest skill is building proofs that check themselves.

CAREER 07 / 16

Data Scientist

The biggest door on this list — and the most crowded



AT A GLANCE

TYPICAL ENTRY	Any quantitative degree plus a portfolio; an MSc in data science or statistics helps
CORE MATHEMATICS	Statistics, probability, linear algebra, optimisation, causal inference
PAY — INDIA	Freshers typically ₹6–9 LPA, with top product firms and AI companies paying well above; overall average around ₹11 LPA (AmbitionBox, 2026)
PAY — ABROAD	US median \$120,230; projected 35% growth 2025–35, among the fastest of any occupation (BLS)
OLYMPIAD EDGE	Medium
WHERE	Technology, e-commerce, banking, healthcare, consulting, government

What the job really is

Data science is sold as a coding job. It is really a statistics job with a coding entrance exam. The tools change every two years; the central question — *is this effect real, or is it noise?* — does not. A data scientist turns messy organisational data into decisions: which product change worked, which customers are about to leave, what the forecast for next quarter should be, and how sure we are.

The objection, taken seriously

The strongest argument against this path is a real one. The field is crowded with bootcamp graduates, many “data scientist” roles are dashboards and reporting, and entry salaries outside top firms are modest. In honesty, all of that is true. But it is exactly where mathematical training separates people: experimental design, causal inference (telling cause from correlation), and the discipline to report uncertainty honestly are rare — and they are what senior roles pay for.

How to get in from India

A BSc or BS in mathematics or statistics, then either direct entry with a strong project portfolio or a specialised master’s (CMI’s MSc in Data Science and ISI’s statistics programmes are examples). Internships matter more than certificates. Learn SQL as seriously as Python.

The catch

Titles are inconsistent — “analyst” and “scientist” can mean the same job or wildly different ones. AI tools are rapidly automating routine analysis, which raises the bar for entry-level work.

START THIS YEAR

- 1 Take one rigorous statistics course and one on causal inference.
- 2 Find a public dataset you care about and write a short report that a non-specialist could act on.
- 3 Learn SQL to the point of writing window functions without looking them up.

Anyone can fit a model. The job is knowing when not to believe it.

CAREER 08 / 16

Actuary

Pricing the future in rupees



AT A GLANCE

TYPICAL ENTRY	Institute of Actuaries of India (IAI) entrance test (ACET) after Class 12, then professional papers alongside work
CORE MATHEMATICS	Probability, statistics, financial mathematics, survival models, stochastic modelling
PAY — INDIA	Roughly ₹4–8 LPA with 0–3 papers; Fellows commonly ₹25–80 LPA (aggregator estimates, indicative)
PAY — ABROAD	US median \$130,000; projected 9% growth 2025–35 (BLS)
OLYMPIAD EDGE	Medium
WHERE	Life and general insurers, reinsurers, pension funds, actuarial consultancies, regulators

What the job really is

An actuary answers questions such as: how much should a thirty-year-old's life insurance cost? How much does a pension fund need to hold today to keep promises due in 2060? What happens to an insurer if a pandemic or a flood arrives next year? Actuaries price risk, set reserves, and advise boards on whether the numbers add up.

Actuarial work is often called the “boring” mathematics career. And yet it is the one where the profession is built to protect you: each exam you pass raises your pay, and the scarcity of qualified actuaries is engineered by the exam system itself.

How to get in from India

You do not need to finish a degree first. The IAI's ACET is open after Class 12, is taken online, and is held twice a year. Passing it makes you a student member; you then clear core technical, core practice and specialist papers to become an Associate (AIAI) and finally a Fellow (FIAI). Many students work while studying. The UK's Institute and Faculty of Actuaries and the US Society of Actuaries offer internationally recognised alternatives.

The catch

The exams take years, often while working full-time, and pass rates are low. Junior work can be routine. The payoff is back-loaded: the career rewards patience more than brilliance.

START THIS YEAR

- 1 Sit ACET early — it is open to Class 12 students.
- 2 Learn R or Python alongside probability; modern actuarial work is computational.
- 3 Clear two or three papers before your first job; entry pay rises sharply with each one.

Actuaries are paid to be careful about the future so that the rest of us can afford not to be.

CAREER 09 / 16

Cryptographer

The number theory underneath every UPI payment



AT A GLANCE	
TYPICAL ENTRY	MS or PhD for research roles; BS for implementation and security engineering
CORE MATHEMATICS	Number theory, abstract algebra, finite fields, lattices, coding theory, complexity theory, probability
PAY — INDIA	Wide; highest at global technology, blockchain and security firms
PAY — ABROAD	US information security analysts: median \$129,180, projected 21% growth (BLS); research cryptographers earn more
OLYMPIAD EDGE	High — number theory and combinatorics are olympiad staples
WHERE	Defence and government research, banks and payment networks, big tech, blockchain companies, security startups, universities

What the job really is

Every secure payment, message and login rests on a mathematical problem nobody knows how to solve quickly. If large quantum computers arrive, some of today’s problems become easy. That is why, in August 2024, the US National Institute of Standards and Technology published its first three post-quantum cryptography standards — FIPS 203, 204 and 205, two of them built on lattices — and in March 2025 selected a further, code-based algorithm (HQC) as a backup. The world now has to migrate its infrastructure, and it is short of people who understand the mathematics.

Cryptography looks like the art of hiding messages. It is really the art of proving that hiding works: a security proof shows that breaking a scheme would require solving a problem believed to be hard.

Four kinds of cryptographer

Designers invent and analyse schemes, usually with a PhD. **Cryptanalysts** try to break them. **Implementers** turn specifications into fast, side-channel-resistant code — most real breaks come from implementation bugs, not mathematics. And **migration specialists** help organisations move to post-quantum systems, the fastest-growing segment right now.

How to get in from India

A mathematics or CS degree with courses in algebra and number theory, then a master’s or PhD for research roles. Capture-the-flag security competitions and the Cryptopals challenges are an excellent bridge between theory and practice. In government, defence research laboratories work on cryptology, though such roles may involve security clearance and citizenship requirements.

The catch

Pure research positions are few and PhD-gated. Government work can be secretive. And the most common way systems fail is not elegant mathematics but careless engineering.

START THIS YEAR

- 1 Take algebra and number theory seriously — groups, rings and finite fields are the working language.
- 2 Work through the Cryptopals challenges in a language you know.
- 3 Read the NIST post-quantum standards' overview sections to see how lattice problems are used.

Mathematics cannot make a system secure. It can only tell you, precisely, what you are trusting.

CAREER 10 / 16

Software Engineer

Mathematics with a compiler attached



AT A GLANCE	
TYPICAL ENTRY	BS; a CS minor or demonstrable projects for mathematics graduates
CORE MATHEMATICS	Discrete mathematics, algorithms, graph theory, logic, probability
PAY — INDIA	Very wide; product companies and top startups pay the most
PAY — ABROAD	US median \$135,980; about 1.7 million jobs (BLS, May 2025)
OLYMPIAD EDGE	Medium to high, through algorithmic interviews
WHERE	Product companies, startups, finance, research labs, every industry

What the job really is

The largest employer of mathematical minds is neither a university nor a bank. It is software. Software engineering is not “coding” any more than writing is “typing”: its core is decomposition, invariants (properties that must always hold), and reasoning about complexity — mathematics with a compiler attached.

How to get in from India

Campus placements, internships and open-source contributions. For mathematics students the gap to close is proof that you can build things that run, not just solve things on paper — a CS minor, a few serious projects, and a competitive-programming record do that. Problem-solving interviews at top companies reward exactly the algorithmic thinking olympiad training builds.

The catch

AI coding tools are reshaping entry-level work quickly. Writing code is becoming cheap; designing systems, finding subtle bugs and guaranteeing correctness are not. That shift favours people with mathematical habits — but only if they also learn to ship.

START THIS YEAR

- 1 Build one complete project end to end — something other people actually use.
- 2 Study algorithms with proofs of correctness, not just implementations.
- 3 Contribute a fix to an open-source project and go through code review.

Code is now the cheapest thing in software to produce. Correct code is not.

CAREER 11 / 16

Statistician & Biostatistician

Making sure the data is not misquoted



AT A GLANCE

TYPICAL ENTRY	MSc in Statistics or M.Stat; Indian Statistical Service through UPSC
CORE MATHEMATICS	Probability theory, inference, regression, design of experiments, Bayesian methods, survival analysis
PAY — INDIA	Indian Statistical Service: Pay Level 10, basic ₹56,100 a month; pharma and research organisations vary widely
PAY — ABROAD	US statisticians: median \$105,650; mathematicians and statisticians projected to grow 10% in 2025–35 (BLS)
OLYMPIAD EDGE	Medium
WHERE	Government statistical system, pharmaceutical companies and clinical research organisations, public health, market research, sports analytics

What the job really is

When a vaccine trial reports a number for efficacy, a statistician designed the experiment that makes that number mean something. Statisticians design surveys and studies, analyse clinical trials, produce official national statistics, and decide what a dataset can and cannot support.

The government route

The Indian Statistical Service is recruited through UPSC's combined IES/ISS examination. The 2026 notification listed 28 ISS vacancies. Eligibility requires a bachelor's degree with statistics, mathematical statistics or applied statistics as a subject, or a master's in one of those fields, and candidates must be 21–30 (with category relaxations). Entry is at Pay Level 10 with a basic of ₹56,100. One warning for mathematics students: a pure mathematics degree without statistics as a subject may not qualify, so read the notification carefully.

The industry route

Pharmaceutical companies and clinical research organisations hire biostatisticians — usually with an MSc in statistics — to design and analyse trials under strict regulatory standards. ISI's B.Stat and M.Stat, and MSc statistics programmes reached through IIT JAM and university entrances, are the main training routes.

The catch

Pay grows more slowly than in technology or finance, and regulated work involves heavy documentation. In return, the work often has direct public consequences.

START THIS YEAR

- 1 Make sure statistics appears on your transcript if the ISS interests you.
- 2 Learn R properly; it remains the standard in many statistical fields.
- 3 Read a published clinical-trial report and try to reconstruct its analysis plan.

Data never speaks for itself. Statisticians are the people paid to make sure it is not misquoted.

CAREER 12 / 16

Operations Research Analyst

The shortest path from a theorem to money saved



AT A GLANCE

TYPICAL ENTRY	BS or MS in mathematics, operations research or industrial engineering
CORE MATHEMATICS	Linear and integer programming, duality, graph theory, queueing theory, simulation
PAY — INDIA	Analytics, logistics and supply-chain roles; wide range
PAY — ABROAD	US median \$88,940; about 113,100 jobs; projected 12% growth 2025–35 (BLS)
OLYMPIAD EDGE	Medium
WHERE	Logistics and e-commerce, airlines, railways, manufacturing, energy, consulting, defence

What the job really is

Airline crew schedules, delivery routes, railway timetables, hospital bed allocation, warehouse layouts — somewhere behind each is an optimisation model. Operations research turns a messy decision into a precise mathematical problem, solves it, and then persuades people to act on the answer.

OR is often treated as applied mathematics' unglamorous cousin. Yet it is the branch with the most direct line from a theorem to money saved, and its methods quietly run much of the modern economy.

How to get in from India

A mathematics degree with optimisation and probability courses, followed by analytics or supply-chain roles, or a master's in operations research or industrial engineering. Fluency in a modelling language and an optimisation solver is as important as the theory.

The catch

The hardest part is rarely the mathematics. It is extracting the real constraints from an organisation and getting a model trusted enough to be used.

START THIS YEAR

- 1 Learn linear programming including duality — it explains why optimal answers look the way they do.
- 2 Model a real scheduling problem (your college timetable will do) and solve it with an open-source solver.
- 3 Study one classic case of optimisation in airlines or logistics.

The optimal solution to the wrong problem is still the wrong answer.

CAREER 13 / 16

Economist & Econometrician

Applied mathematics with people inside it



AT A GLANCE

TYPICAL ENTRY	MA or MSc in economics; PhD for research; Indian Economic Service through UPSC
CORE MATHEMATICS	Real analysis, optimisation, game theory, probability and econometrics
PAY — INDIA	Indian Economic Service starts at Pay Level 10; think tanks, central banking and consulting vary
PAY — ABROAD	US economists: median \$124,720; projected 5% growth 2025–35 (BLS)
OLYMPIAD EDGE	Medium
WHERE	Central banks and finance ministries, research institutes, technology firms, consulting, international organisations

What the job really is

Modern economics is applied mathematics with people inside it. Economists build models of how people, firms and governments respond to incentives, then use *econometrics* — statistics for causal claims — to test those models against data. Technology companies now employ economists to design auctions, set prices and run marketplaces.

How to get in from India

Mathematics graduates transition unusually well into economics, because for most applicants to strong economics master's and PhD programmes, mathematics is the bottleneck. Routes include master's programmes at ISI and leading economics schools in India, PhDs abroad, and the Indian Economic Service, which in 2026 listed 16 vacancies and requires a postgraduate degree in economics, applied or business economics, or econometrics.

The catch

Top research and policy roles are PhD-heavy, and the gap between elegant models and real behaviour is a permanent source of humility.

START THIS YEAR

- 1 Take an introductory microeconomics course alongside real analysis and notice how much is optimisation.
- 2 Learn basic econometrics and replicate a published result from its public data.
- 3 Read a central bank's monetary policy report and trace one argument back to its data.

Economics borrowed mathematics for its rigour. Its best practitioners never forget that people are not equations.

CAREER 14 / 16

Risk Manager & Financial Engineer

Surviving the losses nobody predicted



AT A GLANCE

TYPICAL ENTRY	MSc in mathematics or statistics; master's in financial engineering; FRM or CFA
CORE MATHEMATICS	Stochastic calculus, partial differential equations, Monte Carlo simulation, statistics, numerical methods
PAY — INDIA	Global bank analytics centres in Bengaluru, Mumbai and Gurugram; between data science and trading
PAY — ABROAD	Not tracked separately by BLS
OLYMPIAD EDGE	Medium
WHERE	Investment and commercial banks, asset managers, insurers, rating agencies, regulators

What the job really is

Banks rarely fail from bad mathematics alone. They fail from good mathematics applied to the wrong assumptions — the 2008 crisis was, among other things, a lesson in models trusted beyond their limits. Risk managers measure how much a firm could lose, and financial engineers build and price the instruments that move that risk around.

Risk management is not about predicting losses. It is about surviving the ones nobody predicted.

The mathematics you will actually use

Stochastic calculus — the mathematics of quantities that move randomly and continuously in time — underlies derivatives pricing. Add partial differential equations, Monte Carlo simulation, statistics for credit and market risk, and numerical methods. Model validation, checking other people's models for hidden assumptions, is a whole career in itself.

How to get in from India

Many global banks run large analytics and risk centres in India. Common routes are an MSc in mathematics or statistics followed by campus hiring, a master's in financial engineering abroad, or professional certifications such as the FRM alongside early roles.

The catch

More regulation and less upside than trading, and validation work can feel bureaucratic. But it is a stable career with deep demand for people who understand both the models and their limits.

START THIS YEAR

- 1 Learn measure-theoretic probability before stochastic calculus; it pays off later.
- 2 Price a simple option three ways — formula, binomial tree, Monte Carlo — and compare.
- 3 Read a regulator's report on a past financial failure and list the model assumptions that broke.

A model is a map. Risk management is remembering that the territory gets a vote.

CAREER 15 / 16

Mathematics Educator & Olympiad Coach

The only career on this list that multiplies mathematicians



AT A GLANCE

TYPICAL ENTRY	BSc or MSc plus B.Ed for schools; NET or PhD for colleges; a track record for coaching
CORE MATHEMATICS	Deep command of school and olympiad mathematics — and of explanation itself
PAY — INDIA	Very wide: modest in many schools, far higher for top coaches, authors and education founders
PAY — ABROAD	Varies by country and system
OLYMPIAD EDGE	Very high — credibility is the currency
WHERE	Schools, colleges, olympiad programmes, academies, online platforms, curriculum design

What the job really is

Every IMO medallist had someone who first showed them that a problem could be beautiful. Teaching is often treated as the fallback for mathematics graduates who “couldn’t” do something else. It is closer to the opposite: it is the only career on this list that multiplies mathematicians instead of employing one.

Mathematics educators work in schools and colleges, train olympiad students, write books and problem sets, build online courses, and design curricula. The best of them do something no textbook can — they diagnose how a particular student is thinking.

How to get in from India

School teaching usually requires a B.Ed; college teaching requires NET qualification or a PhD. Olympiad coaching runs on reputation: your own record, your students’ results and the quality of your materials. Online platforms have lowered the barrier to reaching students anywhere in the country.

The catch

Traditional teaching pay has a low ceiling. The coaching market is crowded and often reduces mathematics to drilling — pattern recognition that works up to a point and then stops working. The distinction between training students to repeat and teaching them to think is the whole game, and it is the founding idea behind Liminal.

START THIS YEAR

- 1 Tutor one student for a term and keep notes on where they get stuck — that is your real training.
- 2 Write full solutions to problems for someone else to read; clarity is a skill you can practise.
- 3 Learn how the olympiad ladder works in detail so you can guide students through it honestly.

A good proof convinces a reader. A good teacher produces readers who can write the next proof.

CAREER 16 / 16

Founder

Building companies where mathematics is the product



AT A GLANCE

TYPICAL ENTRY	None — a hard problem, a team, and the willingness to be wrong in public
CORE MATHEMATICS	Whatever the product needs, plus unit economics and probability of survival
PAY	Equity; usually little or nothing at first
OLYMPIAD EDGE	Medium — the network of fellow olympiad alumni is often the real asset
WHERE	Trading, AI, cryptography, optimisation software, education technology

What the job really is

Several of the most talked-about technology companies of the last few years were started by people who first met at mathematics competitions. The AI coding company Cognition was co-founded by olympiad alumni who had interned together at Hudson River Trading; the trading platform Hyperliquid recruits from international olympiad podiums. Mathematical talent plus a network of peers who think the same way is a powerful starting point.

Founders are imagined as people who love risk. The good ones are people who calculate it.

The objection, taken seriously

Most startups fail, and mathematics students are rarely trained in selling, hiring or managing money. That objection is right. The answer is not to avoid founding but to found well: with co-founders whose skills complement yours, a problem you understand better than anyone, and the humility to treat early customers as data.

The catch

No salary for a long time, high odds of failure, and responsibility for other people's livelihoods. It is the most uncertain path here — and the only one where you set the terms of the work.

START THIS YEAR

- 1 Build something small and ship it to real users before you graduate.
- 2 Learn basic accounting and unit economics; they are just arithmetic with consequences.
- 3 Keep in touch with the people you meet at camps and competitions — they are your future co-founders.

A startup is a proof by construction: you do not argue that the thing can exist; you build it.

PART III

Choosing: a plan by stage

Turning the atlas into decisions — what to do at each stage, and the “plus one” each career demands.



PART III · A PLAN BY STAGE

Choosing: a plan by stage

An atlas is useful only if it changes what you do on Monday. What follows is not a script — every student's path bends — but a sequence that keeps the most doors open for the longest time.

Classes 8–10: build the habit

Learn to write a proof, not just reach an answer. Attempt the IOQM early, for the experience rather than the result. Start one programming language. Read mathematics that is not on the syllabus.

Classes 11–12: climb, but do not drill

Take the olympiad ladder seriously, and know the admission routes it opens — CMI, ISI, IIT Bombay, IIT Kanpur, IIT Madras and IIT Kharagpur all reward olympiad performance in different ways. Resist the temptation to replace understanding with volume. Drilling buys pattern recognition that fades; understanding compounds.

Undergraduate years 1–2: the core plus one

Make analysis, algebra, linear algebra and probability rock-solid — every career in this atlas leans on them. Choose your “plus one” early: programming, statistics, economics or finance. Take one course outside your comfort zone each year.

Undergraduate years 3–4: test the waters

Do at least one research project and one industry internship, ideally in different directions. You learn more from discovering what you dislike than from a brochure. Decide between a PhD track (NBHM, CSIR-NET, applications abroad) and industry by the start of your final year.

After the degree: specialise deliberately

Specialisation is where pay and satisfaction diverge, so choose with both in view. A master's or PhD should be a decision, not a default. Whichever path you take, keep your mathematics alive — it is the part of your training that will still be worth something in 2040.

Mathematics plus one

Every career in this atlas pairs mathematics with one complementary skill. Choosing the second skill is, in practice, choosing the career.

Career	The “plus one” it demands
Quant research / trading	Programming and statistical discipline; speed under pressure
Quant development / software	Systems programming and engineering craft
Research mathematics	Patience, exposition and the ability to choose questions
AI / ML research	Deep-learning engineering and experimental judgement
Formal mathematics	Functional programming and logic
Data science / statistics	SQL, communication and causal reasoning
Actuarial / risk	Professional exams and financial knowledge
Cryptography	Secure implementation and computer security
Operations research / economics	Modelling real organisations and institutions
Education / founding	Explanation, empathy and the ability to build teams

Mathematics is not a career. It is the thing that makes all of these careers learnable – and the only one of your skills that is guaranteed not to go out of date.

REFERENCE

Sources and data notes

Figures were checked against the sources below in October 2026. Where sources disagreed, the official or more conservative figure is used. Indian salary figures from aggregators and campus reporting are indicative only.

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Using these pages on the web. Each career section is self-contained — header, at-a-glance panel, five short sections, a “start this year” list and a closing line — so it can be published as its own page with the at-a-glance panel as a sidebar or summary card. Refresh pay and admissions figures each academic year.